

Cost Accounting and Route Consolidation for Pharmaceutical Cold-Chain Logistics

Ruikang Ding[†]

Dian Jin[†]

School of Mathematical Sciences, Fudan University, Shanghai, China

[†]Equal contribution.

Abstract

We develop an operations-research analysis of pharmaceutical cold-chain distribution using dispatch and terminal-delivery records. The study has three connected parts. First, a six-component accounting model augments fixed and mileage costs with refrigeration, utilization penalties, away-from-base waiting, and loading/unloading labor. Applied to 735 dispatch records, the model gives a total of 5.4251 million CNY; mileage-dependent cost accounts for 45.3% and fixed cost for 28.3%. Second, route consolidation is formulated as a capacitated vehicle-routing problem with a detour penalty and solved by a Clarke–Wright initialization followed by a hybrid genetic algorithm and 2-opt local search. Under the project assumptions, the modeled dispatch count falls from 759 to 182 and cost from 5.2088 million to 1.1813 million CNY. Third, terminal operations are assessed through load, punctuality, vehicle matching, carbon emissions, and demand fulfillment. A shipment-ID audit recovers 92,009 cross-file matches and exposes a substantial address-quality problem in the SH9 warehouse data. The paper reports both the optimization gains and the data/model limitations that constrain operational interpretation.

Keywords: pharmaceutical logistics; cold chain; vehicle routing; hybrid genetic algorithm; route consolidation; cost accounting

1 Introduction

Pharmaceutical distribution combines ordinary vehicle-routing decisions with temperature control, service reliability, vehicle compatibility, and unusually high consequences for poor delivery quality. The dataset contains both trunk dispatch records and terminal-delivery records, allowing the analysis to move from accounting to routing and then to operational diagnosis.

The study is not a single monolithic optimization model. It first reconstructs cost at the shipment level, then uses that cost structure to motivate consolidation, and finally checks whether terminal operations exhibit load, service, carbon, or data-quality problems. We preserve that sequence because it reflects the actual research process and makes the assumptions visible.

Classical vehicle routing originates with Dantzig and Ramser’s truck-dispatching formulation, while the Clarke–Wright savings rule remains a standard constructive heuristic.[1, 2] Modern hybrid genetic search combines population-based exploration with local improvement and diversity control for large routing variants.[3] The routing model uses a simpler hybrid scheme tailored to these operational data.

2 Methods

2.1 Cost-Model Overview

For each dispatch record, the accounting model decomposes total cost into six components: fixed vehicle use, mileage-dependent operation, refrigeration, vehicle-utilization penalties, away-from-base waiting, and loading/unloading labor. The first two terms form the transportation baseline, while the remaining four adjust that baseline for cold-chain operation and service conditions.

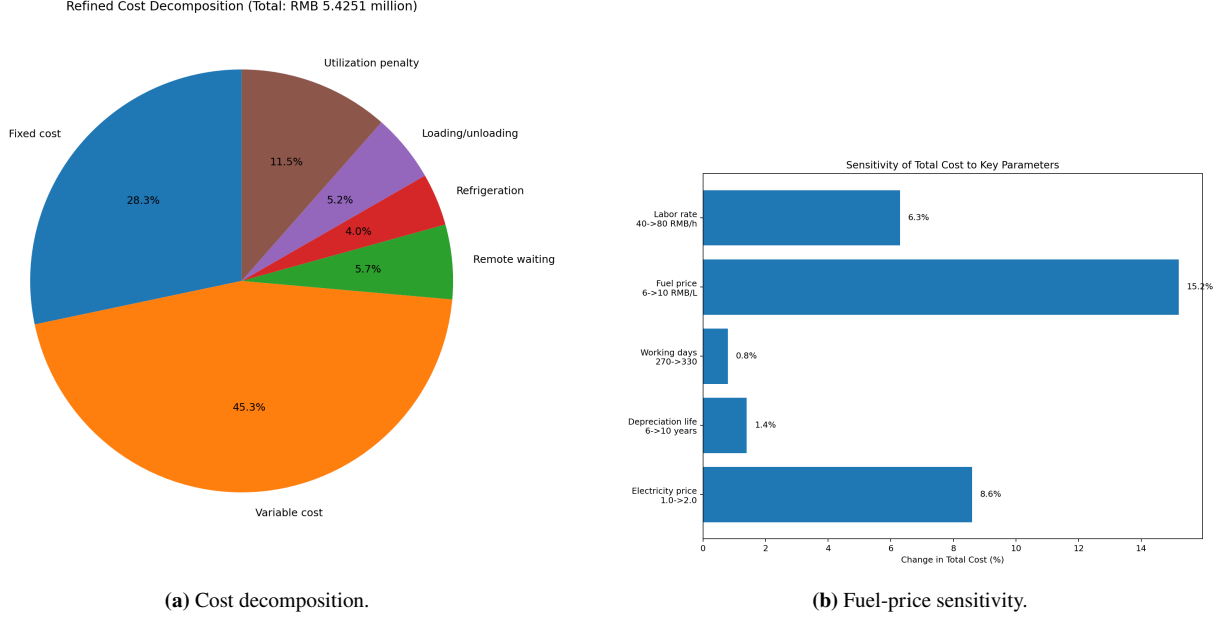
2.2 Fixed and mileage-dependent cost

For vehicle type v , let P_v be purchase price, r residual-value rate, N_{dep} depreciation years, N_{work} annual working days, I_v insurance, A_v inspection cost, R_v road-related annual fee, W_v daily labor, and M_v, F_v monthly violation and maintenance costs. The source model uses straight-line depreciation,

$$C_{fix}^v = \frac{P_v(1-r)}{N_{dep}N_{work}} + \frac{I_v + A_v + R_v}{N_{work}} + W_v + \frac{M_v + F_v}{30}. \quad (1)$$

Table 1: Cost decomposition for 735 dispatch records.

Component	Cost (10,000 CNY)	Share
Fixed vehicle cost	153.39	28.3%
Mileage-dependent cost	245.85	45.3%
Away-from-base waiting	31.18	5.7%
Refrigeration	21.56	4.0%
Loading/unloading	28.28	5.2%
Utilization penalty	62.25	11.5%
Total	542.51	100%

**Figure 1:** Stage-1 cost-accounting diagnostics.

Mileage-dependent cost uses Haversine distance d_h , a route-circuitry factor $\lambda = 1.3$, fuel cost c_{fuel}^v , and toll cost c_{toll}^v :

$$C_{var}^v = (c_{fuel}^v + c_{toll}^v)\lambda d_h. \quad (2)$$

2.3 Cold-chain and operating adjustments

Four additional terms are introduced. Refrigeration is approximated by a 5.0 kW unit and electricity price 1.5 CNY/kWh, so $C_{ref} = 5.0T_{hours} \times 1.5$. Loading and unloading uses two workers, two hours, and 50 CNY/h, giving 200 CNY per operation. For trips beyond 1000 km, away-from-base waiting is modeled as $0.3C_{fix}t$.

Vehicle utilization $\eta = q/Q_v$ is penalized with a piecewise quadratic function,

$$C_{pen}^v(\eta) = \begin{cases} 2C_{fix}^v(0.5 - \eta)^2, & \eta < 0.5, \\ 0, & 0.5 \leq \eta \leq 1, \\ 0.4C_{var}^v(\eta - 1)^2, & \eta > 1. \end{cases} \quad (3)$$

The analysis explicitly checks continuity of both the function and first derivative at $\eta = 0.5$ and $\eta = 1$. This smoothing was introduced after an earlier step-like penalty caused unstable behavior near the thresholds.

For dispatch record j , total modeled cost is

$$C_{total}^{(j)} = C_{fix}^{vj}t_j + C_{var}^{vj}(d_j) + C_{ref}^{vj} + C_{pen}^{vj}(\eta_j) + C_{wait}^{vj}(t_j) + n_jC_{lu}. \quad (4)$$

Mileage and fixed dispatch cost together account for 73.6% of the modeled total. This concentration is what makes route consolidation economically consequential in the next stage.

Table 2: Detour-parameter experiment reported in the source project.

α	Cost (10,000 CNY)	Vehicles	Saving	Implausible merges
1.2	128.5	210	75.3%	0
1.3	122.3	195	76.5%	1
1.4	118.13	182	77.3%	3
1.5	118.5	180	77.2%	8
1.6	120.2	178	76.9%	15
1.8	124.8	175	76.0%	35
2.0	130.1	172	75.0%	55

3 Results

3.1 Sensitivity Analysis

For parameter p , the project defines a local sensitivity coefficient

$$S_p = \frac{\Delta C / C_0}{\Delta p / p_0}. \quad (5)$$

Fuel price, electricity price, labor rate, depreciation period, and working days are varied. The largest reported effect comes from fuel: increasing fuel price from 6 to 10 CNY/L raises total modeled cost by 15.2%, with $S_{oil} = 0.76$. Electricity and labor are reported as moderate ($S \approx 0.15$), while depreciation period and working days have $|S| < 0.05$.

The analysis is deliberately one-factor-at-a-time. It identifies which assumptions deserve attention, but it does not represent joint uncertainty in fuel, labor, electricity, and demand.

Route-Consolidation Model. For each day d , let N_d be the customer set, node 0 the depot, and K_d the vehicle set. Binary x_{ijk} indicates whether vehicle k travels from i to j , y_{ik} assigns customer i to vehicle k , and u_{ik} is an MTZ auxiliary variable. The objective combines travel cost, fixed vehicle cost, and a detour penalty,

$$\min Z = \sum_{k,i,j} c_{ij}^k x_{ijk} + \sum_k C_{fix}^{v_k} t_k + \sum_k \alpha s_k. \quad (6)$$

The main constraints require each customer to be served once, vehicle load not to exceed capacity, at most four stops per route, flow conservation, and MTZ subtour elimination. The stop limit is a proxy for the 09:00–17:00 service window because the source dispatch file does not contain enough intra-day timing information for an exact VRPTW formulation.

A route-distance condition compares consolidated distance D_k with the sum of independent depot round trips D_k^{ind} . The project studies $\alpha \in \{1.2, 1.3, 1.4, 1.5, 1.6, 1.8, 2.0\}$ and uses 1.4 as the working setting.

The table explains the choice more clearly than a generic claim of optimality. Tight settings reject useful nearby combinations; looser settings permit increasingly implausible long detours. The minimum reported cost occurs at 1.4, while 1.5 is nearly tied but produces more questionable merges.

Hybrid Genetic Algorithm. The constructive stage uses Clarke–Wright savings

$$S_{ij} = d_{0i} + d_{0j} - d_{ij}, \quad (7)$$

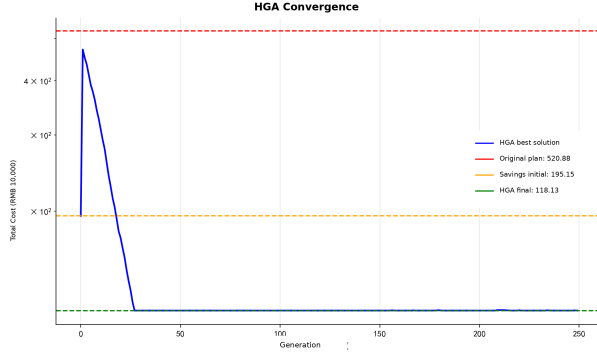
and greedily merges positive-saving pairs subject to capacity, a four-stop limit, a 500 km pair-distance screen, and the detour rule. This produces a structured starting solution rather than a random population.

The genetic stage uses integer-permutation chromosomes with route separators. Eighty percent of the initial population is generated by perturbing savings solutions and 20% is random. Tournament selection has size 3; order crossover uses probability 0.85; swap, insertion, and inversion mutations share a total mutation probability of 0.15; each decoded route is improved with 2-opt; and the top 10% of a population of 120 are retained. The final setting uses 250 generations.

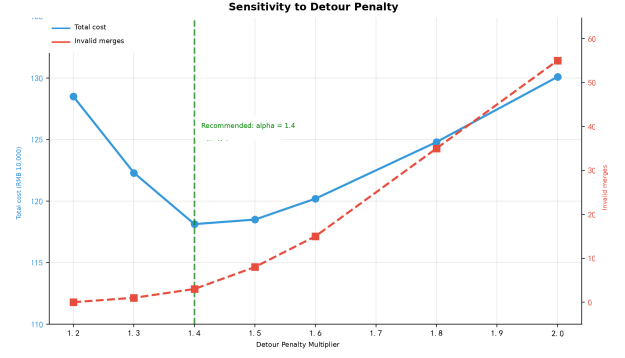
These parameters were not chosen in one step. The analysis records that an early population of 50 and 100 generations gave a 20-run coefficient of variation of 8.3%. Increasing the population to 120 and the horizon to 250 generations reduced that figure to 1.79%, at the cost of longer runtime. Raising crossover from 0.70 to 0.85 improved solution quality, whereas 0.95 reduced diversity and caused premature convergence. This tuning history is useful because it explains why the final parameter set is a compromise rather than a theoretically privileged choice.

The reported per-generation complexity is dominated by 2-opt and scales as $O(N_{pop}|N_d|^2)$. Across G_{max} generations, the analysis gives the overall order as

$$O(G_{max}N_{pop}|N_d|^2). \quad (8)$$



(a) HGA convergence.



(b) Detour-penalty sensitivity.

Figure 2: Optimization behavior recorded in the original route-consolidation study.

Table 3: Baseline and HGA solution under the project model.

Plan	Cost (10,000 CNY)	Vehicles	Saving
Single-destination baseline	520.88	759	—
HGA consolidated plan	118.13	182	77.3%

For a typical day with about 100 customer nodes, the report gives a Python runtime of roughly 5–8 minutes on an Intel i7-10700 for the stated implementation. A retrospective paragraph notes that the expanded configuration could take about 15 minutes during tuning. We retain both statements as implementation-dependent observations rather than forcing a single runtime.

Consolidation Results. The modeled cost falls by 4.0275 million CNY and the dispatch count by 577. The report attributes about 60% of the saving to Shanghai, Jiangsu, and Zhejiang, where high customer density makes consolidation easier. Twenty independent runs give a 1.79% coefficient of variation.

This 77.3% reduction is a model comparison, not an observed post-deployment saving. The baseline assumes single-destination dispatch and the optimized model simplifies time windows, vehicle heterogeneity, and road-network travel. The result nevertheless demonstrates the scale of consolidation potential under the stated assumptions.

Terminal-Distribution Evaluation. The third stage evaluates five dimensions: load factor, punctuality, vehicle-type matching, carbon emissions, and demand fulfillment. For load q_m and nominal capacity Q_{v_m} , $\eta_m = q_m/Q_{v_m}$. Punctuality is defined by a 96-hour threshold between dispatch and receipt. Carbon emissions are calculated from distance, vehicle fuel consumption, and an emission factor of 2.68 kg CO₂/L.

For 222,378 terminal records, the report gives an average load factor of 143.60%, a 17.5% share below 50% load, 99.16% punctuality, a seven-hour average transport duration, and 157.73 t CO₂ total modeled emissions. The load figure above 100% is not treated as a virtue: the report itself diagnoses simultaneous overload and underload, indicating either genuine capacity violations or a mismatch between recorded shipment units and nominal capacity.

4 Additional Model Diagnostics and Data Analysis

4.1 Address Audit and the SH9 Case

A major part of the research process was triggered by a data failure rather than by the optimization algorithm. In the terminal file, 33,302 SH9 records share the same receiving address, causing a naive geocoding pipeline to collapse them to one point and produce the absurd result of a one-vehicle solution.

Shipment identifiers were then matched across the two operational tables. The audit found 92,009 common IDs overall. For SH9, 13,580 of 33,302 records match, a 40.8% rate, covering 1,823 unique recovered addresses. The report states that addresses for the matched IDs agree 100% across the two files. Of those unique SH9 addresses, 977 are in Shanghai and 846 are outside Shanghai.

For the matched subset, the project applies a greedy nearest-neighbor VRP with nominal capacity 100 items and maximum service radius 80 km. The reported SH9 requirement for this track is about 400 vehicles. The remaining 19,722 unmatched records are extrapolated from the observed address density, initially producing about 600 additional vehicles. After assuming gains from same-area consolidation, dynamic vehicle matching, and time-window clustering,

Table 4: Selected terminal-distribution indicators.

Indicator	Reported value
Average load factor	143.60%
Share below 50% load	17.5%
On-time rate	99.16%
Average transport duration	7.0 h
Modeled carbon emissions	157.73 t CO ₂

the report gives a practical weekly range of 650–720 vehicles.

The detailed SH9 calculation yields a practical weekly planning range of 650–720 vehicles after same-area consolidation, dynamic vehicle matching, and time-window clustering. This range is used consistently in the analysis below.

Parameter Sensitivity and Cost Drivers. The source project defines a dimensionless one-factor sensitivity coefficient

$$S_p = \frac{\Delta C / C_0}{\Delta p / p_0}, \quad (9)$$

where C_0 is baseline total cost and p_0 is the baseline parameter. Fuel price, electricity price, labor rate, depreciation life, and annual working days are varied around their baselines. The strongest reported effect is fuel: moving from 6 to 10 CNY/L increases total modeled cost by 15.2%, corresponding to $S_{oil} = 0.76$. Electricity and labor are reported near 0.15, while depreciation life and working days have absolute sensitivity below 0.05.

This ordering is consistent with the cost decomposition. Mileage-dependent transport already contributes 45.3% of baseline cost, so a parameter that enters that term broadly across routes has greater leverage than a parameter affecting only refrigeration or fixed-cost allocation. The sensitivity exercise therefore motivates the second stage of the project: reducing unnecessary mileage and dispatches is likely to matter more than fine-tuning a minor accounting coefficient.

Smooth Utilization Penalty. A distinctive modeling choice in the analysis is the use of a smooth quadratic penalty rather than a step penalty for low loading and overload. For load factor $\eta = q/Q_v$,

$$C_{pen}^v(\eta) = \begin{cases} 2C_{fix}^v(0.5 - \eta)^2, & \eta < 0.5, \\ 0, & 0.5 \leq \eta \leq 1, \\ 0.4C_{var}^v(\eta - 1)^2, & \eta > 1. \end{cases} \quad (10)$$

At $\eta = 0.5$, both the left-hand function value and derivative equal zero, matching the middle branch. At $\eta = 1$, the right-hand function value and derivative also equal zero. The penalty is therefore C^1 at both thresholds under the formula used in the project.

The report records that an earlier stepwise design created an abrupt fitness landscape near the utilization thresholds. A linear transition reduced the jump in function value but retained derivative discontinuities. The quadratic form was adopted after these trials. Although a genetic algorithm does not require gradients in the same way as a gradient-based optimizer, a smoother objective can still reduce abrupt changes in fitness under small route perturbations.

Detour-Constraint Rationale. The detour term was introduced after an early version of the optimizer found combinations that were mathematically feasible but operationally implausible. The analysis gives the example of combining Shanghai, Beijing, and Guangzhou in one route, creating nearly 2,000 km of additional travel. The fixed-cost saving of roughly 685 CNY was much smaller than the roughly 4,000 CNY increase in variable cost. This failure motivated an explicit restriction on route elongation.

The subsequent sweep from $\alpha = 1.2$ to 2.0 shows the expected trade-off. At 1.2 the rule is so strict that useful nearby combinations are lost; at 1.6 and above, questionable merges rise quickly and variable cost begins to offset the benefit of using fewer vehicles. The chosen 1.4 setting is therefore supported by a discrete experiment in the source project rather than selected solely by intuition.

Algorithmic Workflow. The HGA can be viewed as a sequence of constructive, population-based, and local-search operations. Clarke–Wright savings first supplies routes with sensible geographic structure. Encoding those routes as chromosomes allows crossover and mutation to explore alternative groupings. After decoding, 2-opt removes avoidable crossing edges inside each route. Elitism then protects the best solutions while tournament selection maintains competitive pressure.

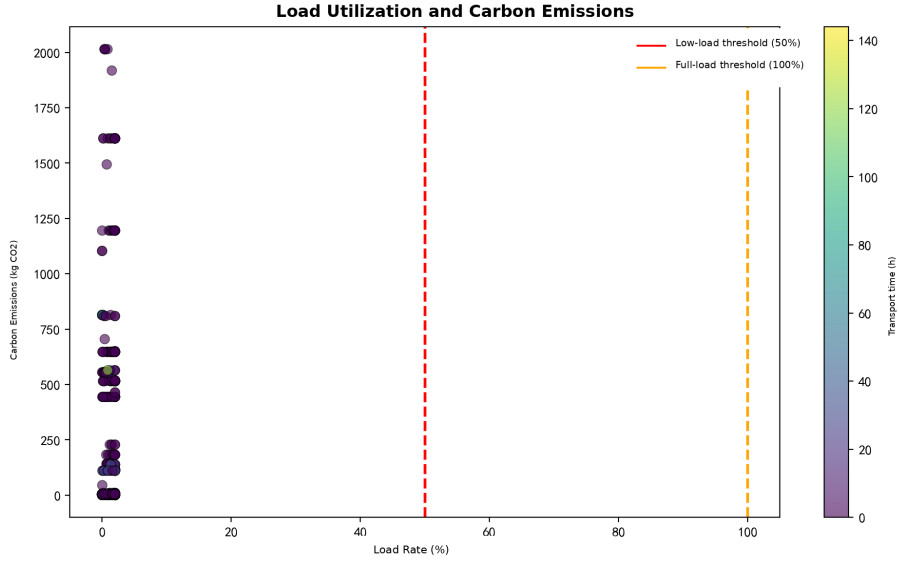


Figure 3: Terminal-distribution diagnostics for load utilization and carbon-related performance.

The source pseudocode evaluates every decoded individual, applies local search, retains the top 10%, and fills the remaining population through tournament-selected parents, order crossover, and mutation. The reported complexity $O(G_{max} N_{pop} |N_d|^2)$ reflects the repeated quadratic local-search stage. This is consistent with the broader vehicle-routing literature, where hybrid genetic search commonly combines population exploration with aggressive route improvement.[3]

The tuning history is also informative. A population of 50 and 100 generations initially produced a 20-run coefficient of variation of 8.3%. Increasing the population to 120 and generations to 250 reduced it to 1.79%. Crossover 0.85 improved solution quality relative to 0.70, whereas 0.95 led to premature convergence. The final parameters therefore reflect an empirical balance among diversity, convergence, and runtime.

Terminal Evaluation Metrics. The third stage deliberately moves away from a single cost objective. Load factor measures asset utilization but also exposes overload; punctuality measures service; vehicle-type matching checks whether assigned capacity is appropriate; carbon accounting converts fuel use to emissions; and demand fulfillment measures whether planned capacity actually covers orders. These indicators can disagree, which is precisely why they are reported together.

The source project estimates that consolidating low-load vehicles below 50% with nearby routes could reduce 15–20% of ineffective mileage, corresponding to 23.7–31.5 t CO₂ under its emissions assumptions. This is an engineering estimate, not an observed reduction. It depends on whether those orders are geographically and temporally compatible, a condition that the aggregate load statistic alone cannot establish.

The average load factor of 143.60% is particularly important. If shipment quantity and nominal vehicle capacity are measured on compatible scales, the figure indicates systematic overload. If they are not, it identifies a data-definition problem. Either interpretation requires investigation before using load factor as an optimization target.

SH9 Demand Reconstruction. The SH9 case uses two tracks because only 40.8% of its records can be linked to a second file with recoverable addresses. Track 1 operates on 13,580 matched records and 1,823 unique addresses. Of these addresses, 977 are located in Shanghai and 846 elsewhere. The project applies a greedy nearest-neighbor routing procedure with capacity 100 items and an 80 km service-radius rule, obtaining roughly 400 vehicles for the matched subset.

Track 2 extrapolates the unmatched records from the address density observed in Track 1. The detailed section states that 13,580 records cover 1,823 unique addresses, about 7.45 records per address, and uses this ratio to estimate roughly 2,650 addresses for the unmatched portion. Applying the same routing logic gives about 600 additional vehicles before operational consolidation assumptions. Same-district merging, dynamic vehicle sizing, and time-window clustering then reduce the reported practical range to 650–720 vehicles per week.

The extrapolation is the weakest evidential step in this part of the project. The unmatched 60% need not have the same spatial distribution as the matched 40%. The result is therefore best used as a capacity-planning range rather than a precise fleet requirement. The analysis itself notes that complete de-identified addresses would materially improve the estimate.

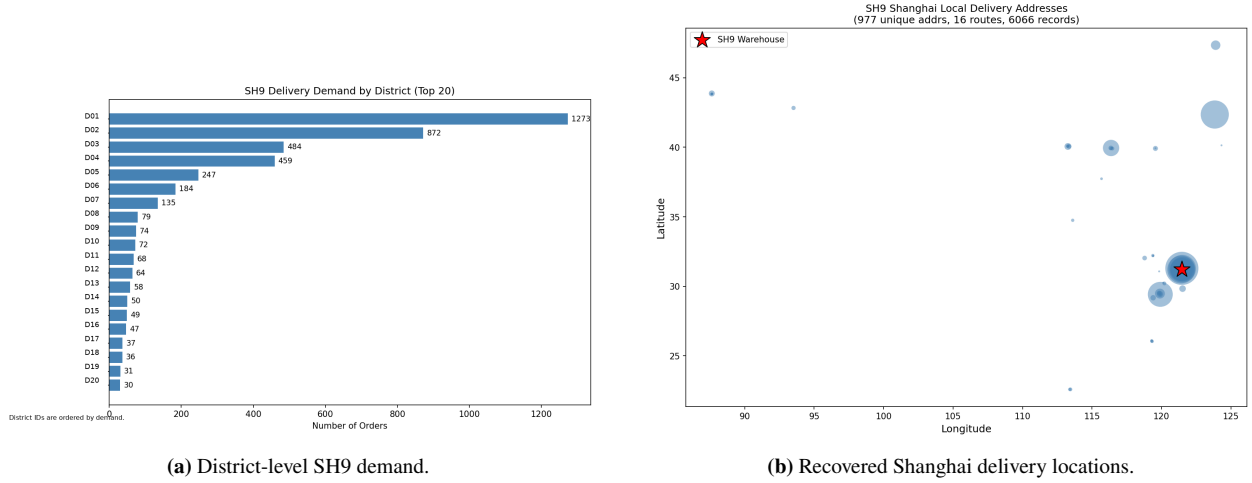


Figure 4: Spatial diagnostics used in the SH9 address-recovery analysis.

Cost Example. The appendix gives a concrete Shanghai-to-Nanjing refrigerated shipment with 200 items. The fixed-cost allocation is 685.00 CNY; variable transport is 780.00 CNY using the project’s distance and road-factor assumptions; refrigeration is 28.13 CNY; utilization penalty is zero at an 85% load factor; and one loading/unloading operation costs 200.00 CNY. The modeled total is therefore 1,693.13 CNY.

This example is useful because it shows how the six-component accounting model is actually assembled at record level before aggregation over 735 dispatch records. It also makes clear why a route change can affect several cost terms at once: combining orders can reduce fixed dispatch cost and mileage, but may increase refrigeration time, loading operations, or detour cost.

5 Discussion

5.1 Limitations and Research Lessons

Several limitations are substantive. The time-window constraint is represented by a four-stop cap because timestamps are incomplete. Refrigeration power, waiting-cost fraction, and some operating thresholds are industry-style assumptions rather than measurements from the focal company. The SH9 reconstruction observes only 40.8% of records directly; the remainder is extrapolated. The large modeled saving therefore should not be interpreted as an implementation guarantee.

The development history also contains a useful warning about small-sample inspection. An early comparison of 100 rows found no common shipment IDs across two files and suggested that the coding systems were incompatible. A later full-file audit found 92,009 common IDs. The correction materially changed the terminal-routing analysis. In this project, data linkage turned out to be at least as important as the routing heuristic.

A deployment study would replace Haversine distance with road-network travel times, use complete geocoded destinations, enforce explicit time windows and driver hours, model weight and volume jointly, and validate cost coefficients against audited operational data. Multiobjective routing could then trade off cost, carbon, and service rather than treating them in separate stages.

Constraint Set and Feasibility Interpretation. The routing formulation in the analysis contains five operational constraint groups. Each customer is assigned to exactly one vehicle. Total assigned demand cannot exceed the selected vehicle capacity. Each route is limited to four customer stops as a proxy for the missing intra-day time-window information. Flow-conservation constraints make arrival and departure consistent with customer assignment, and MTZ auxiliary variables remove disconnected subtours.

The detour rule compares the actual consolidated distance D_k with the sum of independent depot round trips D_k^{ind} . In the report this is written with a big- M auxiliary form and the working coefficient $\alpha = 1.4$. Conceptually, the rule prevents a reduction in vehicle count from being purchased with an arbitrarily long route. This is especially important because the objective already rewards consolidation through fixed-cost savings.

Feasibility in this model is therefore operationally approximate. Capacity and route connectivity are explicit, but service time is represented by stop count rather than arrival time. An exact arrival-time model is not used because the dispatch records do not provide timestamps at sufficient resolution. The four-stop cap should therefore be interpreted as an operational approximation rather than as a guarantee that every four-stop route satisfies a 09:00–17:00 window.

Greedy VRP Used in the Address-Recovery Stage. The terminal-address analysis uses a simpler routing routine than the trunk HGA. Starting from the depot, it repeatedly selects the nearest unserved address that remains within the 80 km service radius and does not violate the nominal 100-item capacity. When no further address can be added, the route is closed and a new vehicle starts from the depot. This continues until the recovered-address set is exhausted.

The algorithm is intentionally transparent because the address audit, not metaheuristic sophistication, is the main uncertainty in this stage. A more advanced optimizer would not fix the fact that roughly 60% of SH9 records cannot be linked to a recoverable address. The project therefore uses the greedy result as a capacity estimate for the observed subset and treats the remainder by interpolation.

The same logic is applied to several warehouses in the detailed report. SH9 has 33,302 records, 13,580 linked records, and 1,823 unique recovered addresses, with roughly 400 vehicles for the matched subset. The source table also gives approximate matched-subset estimates of 340 vehicles for XJ1, 195 for ZZ1, 245 for GZ1, and 310 for TY3. These numbers are not combined into a system-wide fleet claim because address coverage and geography differ substantially across warehouses.

Operational Implications. The analysis proposes four practical directions. First, orders in the same administrative district or within roughly 15 km can be candidates for consolidation. Second, vehicle size can be matched dynamically to order volume, with small vans for 0–50 items, small trucks for 50–150, and standard vehicles for 150–300 in the illustrative operating rule. Third, orders can be clustered by requested delivery period to reduce waiting and empty mileage. Fourth, a large urban node such as SH9 could separate trunk movement from terminal distribution through forward depots.

These recommendations should be read as hypotheses generated by the model. These recommendations are model-based rather than outcomes of a field deployment. In particular, expected vehicle-count reductions and load-factor improvements depend on complete addresses, compatible delivery windows, and actual vehicle capacities. Accordingly, projected percentages are interpreted as model-based operational estimates rather than observed field outcomes.

Carbon Accounting and the Cost–Service Trade-off. Carbon emissions are estimated from travel distance, vehicle fuel consumption, and a factor of 2.68 kg CO₂/L. This makes emissions mechanically responsive to route length and vehicle use, so many cost-saving consolidations also reduce modeled carbon. The objectives are not identical, however. A route that reduces the number of dispatches can still create a long detour, and a highly loaded route can violate capacity or service constraints.

This is why the source project reports carbon alongside load factor, punctuality, vehicle matching, and fulfillment rather than folding every criterion into the monetary objective. A future multiobjective formulation could make the trade-off explicit, but the present analysis is better interpreted as a cost optimization followed by operational and environmental diagnostics.

Assumptions and Cost-Parameter Provenance. The cost model relies on several explicit engineering assumptions. Road distance is approximated by Haversine distance multiplied by a curvature factor of 1.3. Refrigeration power is set to 5.0 kW, motivated in the analysis by representative refrigerated-unit nameplate ranges, and electricity is valued at 1.5 CNY/kWh. Loading and unloading use two workers for two hours at 50 CNY per worker-hour, giving 200 CNY per operation. For long trips above 1,000 km, the model permits a wait-versus-return decision and values away-from-base waiting at 30% of daily fixed vehicle cost.

These quantities are not measurements from the focal logistics company. The analysis explicitly classifies several of them as industry-typical estimates. Their role is to make the accounting model operational enough for sensitivity analysis, not to provide audited financial statements. This distinction is why the paper reports the sensitivity coefficients and avoids treating the 5.4251 million CNY baseline as an observed ledger total.

The model also assumes that each dispatch record can be evaluated independently in the accounting stage, that vehicle capacity is represented by the available nominal-capacity field, and that refrigeration runs throughout the modeled transport time. These simplifications are reasonable for a first cost reconstruction but would need refinement if detailed telematics, temperature logs, driver shifts, and actual toll records were available.

Repeated-Run Robustness. The 20-run coefficient of variation of 1.79% is the main stochastic robustness statistic reported for the HGA. It is useful because genetic algorithms can return different solutions under different initial populations and mutation paths. The source tuning history shows that this stability was not automatic: the earlier population-50, generation-100 configuration had a reported coefficient of variation of 8.3%.

The improvement after increasing population size and search horizon indicates that the final configuration explores the solution space more consistently, although it also raises runtime. The report describes this as a practical compromise: enough diversity to avoid frequent poor local solutions, but not an attempt to prove optimality. This is consistent with the role of hybrid genetic search in difficult vehicle-routing problems, where computational studies typically emphasize repeated solution quality and robustness rather than exact certificates on large instances.[3]

The detour-parameter sweep provides a second robustness dimension. Instead of reporting only the selected $\alpha = 1.4$, the source shows how cost, vehicle count, saving rate, and implausible merges evolve from 1.2 through 2.0. The selected setting sits near the bottom of the reported cost curve before implausible merges rise sharply. Together, the repeated-run test and parameter sweep give stronger support than either one alone.

6 Conclusion

The project connects detailed cost accounting, route consolidation, algorithm tuning, terminal-service evaluation, and data-quality auditing. Mileage and fixed dispatch costs dominate the modeled baseline. Under the chosen CVRP-RP assumptions, the HGA reduces modeled cost from 5.2088 to 1.1813 million CNY and dispatches from 759 to 182, with stable repeated-run performance. The terminal analysis, however, prevents an overly simple conclusion: overload indicators, incomplete addresses, and extrapolated SH9 demand remain material uncertainties. The main result is therefore not only a low-cost route plan, but a documented chain from cost drivers to optimization and then to the operational checks needed before deployment.

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